

3.2.8 H: Hyperbolic functions

	Content
H1	Understand the definitions of hyperbolic functions $\sinh x$, $\cosh x$ and $\tanh x$, including their domains and ranges, and be able to sketch their graphs. Understand the definitions of hyperbolic functions $\operatorname{sech} x$, $\operatorname{cosech} x$ and $\coth x$, including their domains and ranges.
	Content
H2	Differentiate and integrate hyperbolic functions.
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H3	Understand and be able to use the definitions of the inverse hyperbolic functions and their domains and ranges.
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H4	Derive and use the logarithmic forms of the inverse hyperbolic functions.
	Content
H5	Integrate functions of the form $(x^2 + a^2)^{-\frac{1}{2}}$ and $(x^2 - a^2)^{-\frac{1}{2}}$ and be able to choose substitutions to integrate associated functions.

	Understand and use $\cosh^2 x - \sinh^2 x \equiv 1$; $\operatorname{sech}^2 x \equiv 1 - \tanh^2 x$ and $\operatorname{cosech}^2 x \equiv \coth^2 x - 1$, $\cosh 2x \equiv \cosh^2 x + \sinh^2 x$, $\sinh 2x \equiv 2\sinh x \cosh x$
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	Content
H7	Construct proofs involving hyperbolic functions and identities.